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May 10, 2026

Hiring Manager

Upstart

Dear Hiring Manager,

My name is Ajay Venkatesh — finishing my M.S. in Computer Engineering at NYU, with several years of production software experience: a few years at PwC in Bengaluru, a summer SDE internship at Amazon, and ongoing on-campus work at NYU's Novel AI Technologies. My focus has been on **full-stack product systems, distributed backends**, and the practical **AI layer** — LLM-powered features, RAG pipelines, and agentic workflows that ship in real products.

The Upstart Borrower Experience posting hits a few of my strongest threads. At NYU's Novel AI Technologies I shipped a customer-facing insurance analytics platform end-to-end — **React, Node.js, REST APIs, Docker on EC2**, with Stripe billing, role-based access, and row-level security across paying clients. The shape of that work — borrower-style self-service, financial workflows, and reducing friction in the user journey — maps directly onto what the posting describes. I also led a broker-portal integration with insurer platforms using **document/media processing** and **WebSocket-based real-time communication** — the same kind of "translate operational workflows into intuitive product experiences" the role calls out.

On AI: at Amazon I deployed a **Retrieval-Augmented Generation (RAG)** pipeline with an **agentic workflow** that autonomously retrieves operational logs and generates context-aware summaries and remediation steps. I implemented an **MCP server** for tool-augmented retrieval and built a high-throughput, event-driven triage system on **ECS Fargate, SQS, SNS** that materially reduced response latency. At Campus Mesh I built **CampusIQ**, an LLM-powered AI assistant with **prompt engineering** and **RAG retrieval** over structured data, with embedding-based semantic search to improve contextual accuracy. Building LLM-powered features in user-facing surfaces, with safety and explainability in mind, is exactly where I want to spend the next chapter.

Stack-wise: **Python** and **TypeScript / JavaScript** for backend and product surfaces; **React** for web; cloud microservices on **AWS** with **Docker** and CI/CD; **SQL** (PostgreSQL, MySQL, DynamoDB) for the data layer. I'm comfortable spanning the stack, partnering with product / design / analytics, and reasoning about reliability and safety in customer-facing systems.

What pulls me toward Upstart specifically is the mission — reducing the cost and complexity of borrowing through AI — and the fact that the work is measured by real outcomes for real borrowers, not internal vanity metrics. That's the kind of feedback loop I want to be inside of.

I'd welcome the chance to discuss how I could contribute to the Borrower Experience team. Thanks for considering my application.

Sincerely,

Ajay Venkatesh